

CHAPTER 1

Data Tables

1.1 Complete Dataset

Table 1.1 presents the complete dataset used in this study as obtained from the World Bank World Development Indicators database online.

Table 1.1: Complete Ghana Dataset: Tax Rate and GDP Growth (2001–2023)

Year	τ (%)	τ (decimal)	τ^2	g (%)
2001	17.1929	0.1719	0.029559	4.0000
2002	17.4931	0.1749	0.030601	4.5000
2003	18.4777	0.1848	0.034142	5.2000
2004	21.7521	0.2175	0.047316	5.6000
2005	21.3215	0.2132	0.045461	5.9000
2006	12.5346	0.1253	0.015712	6.3999
2007	13.8780	0.1388	0.019266	4.3468
2008	13.8960	0.1390	0.019310	9.1498
2009	12.6125	0.1261	0.015908	4.8445
2010	13.3884	0.1339	0.017925	7.8997
2011	14.8658	0.1487	0.022099	14.0471
2012	15.3682	0.1537	0.023618	9.2928

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Table 1.1 — continued

Year	τ (%)	τ (decimal)	τ^2	g (%)
2013	10.6718	0.1067	0.011389	7.3125
2014	11.2522	0.1125	0.012661	2.8562
2015	11.6906	0.1169	0.013673	2.1208
2016	11.0583	0.1106	0.012229	3.3735
2017	11.5770	0.1158	0.013409	8.1289
2018	12.2442	0.1224	0.014992	6.2001
2019	11.9970	0.1200	0.014393	6.5078
2020	11.3404	0.1134	0.012861	0.5139
2021	12.2447	0.1224	0.014993	5.0765
2022	12.2975	0.1230	0.015123	3.8048
2023	12.3931	0.1239	0.015359	3.1401
Mean	13.9803	0.1398	0.019878	5.6616
Std Dev	3.1941	0.0319	0.009774	2.8613
Min	10.6718	0.1067	0.011389	0.5139
Max	21.7521	0.2175	0.047316	14.0471

1.2 Fitted Values and Residuals

Table 1.2 shows the fitted values and residuals from the estimated quadratic model.

Table 1.2: Actual vs. Fitted GDP Growth Rates and Residuals (Ghana, 2001–2023)

Year	Actual g (%)	Fitted $\hat{g}$ (%)	Residual e (%)
2001	4.0000	6.1452	−2.1452
2002	4.5000	6.2421	−1.7421
2003	5.2000	6.3922	−1.1922
2004	5.6000	5.4061	+0.1939
2005	5.9000	5.5755	+0.3245
2006	6.3999	6.6733	−0.2734
2007	4.3468	6.7461	−2.3993
2008	9.1498	6.7370	+2.4128
2009	4.8445	6.6568	−1.8123
2010	7.8997	6.7713	+1.1284
2011	14.0471	6.9278	+7.1193
2012	9.2928	6.8793	+2.4135
2013	7.3125	5.8444	+1.4681
2014	2.8562	6.0638	−3.2076
2015	2.1208	6.2069	−4.0861
2016	3.3735	5.9820	−2.6085
2017	8.1289	6.1645	+1.9644
2018	6.2001	6.4946	−0.2945
2019	6.5078	6.4001	+0.1077
2020	0.5139	6.1244	−5.6105
2021	5.0765	6.4948	−1.4183
2022	3.8048	6.4786	−2.6738
2023	3.1401	6.4488	−3.3087

Year	Actual g (%)	Fitted $\hat{g}$ (%)	Residual e (%)
SSE			205.43
R^2			0.1443